

# OFFICIAL HANDBOOK

*PROJECT NEXUS 2026:  
AI INNOVATION CHALLENGE*



Organised By:

**IEEE MMU SB 2026**

Institute of Electrical & Electronics Engineers  
Multimedia University Student Branch  
2026



Multimedia University (Cyberjaya Campus)  
Persiaran Multimedia, 63100 Cyberjaya, Selangor

## **TABLE OF CONTENTS**

|                                                             |          |
|-------------------------------------------------------------|----------|
| <b>1. Eligibility.....</b>                                  | <b>1</b> |
| 1.1. Who May Participate.....                               | 1        |
| 1.2. Team Composition.....                                  | 1        |
| 1.3. Women in Engineering (WIE) Requirement.....            | 1        |
| 1.4. Registration Fees.....                                 | 1        |
| <b>2. Competition Tracks.....</b>                           | <b>2</b> |
| 2.1. Track 1 — Healthcare Technology.....                   | 2        |
| 2.2. Track 2 — Sustainable Solutions.....                   | 2        |
| 2.3. Track 3 — Global Impact.....                           | 2        |
| <b>3. Phase 1 — Registration &amp; Ideation.....</b>        | <b>3</b> |
| 3.1. Key Dates.....                                         | 3        |
| 3.2. Proposal Submission Requirements.....                  | 3        |
| 3.3. Phase 1 Scoring Criteria.....                          | 3        |
| <b>4. Phase 2 — Selection, Funding &amp; Workshops.....</b> | <b>4</b> |
| 4.1. Seed Grant.....                                        | 4        |
| 4.2. Workshops & Talks.....                                 | 4        |
| <b>5. Phase 3 — Build &amp; Mentorship.....</b>             | <b>5</b> |
| 5.1. Build Period.....                                      | 5        |
| 5.2. Mentorship.....                                        | 5        |
| 5.3. Grand Finale Selection.....                            | 5        |
| 5.4. Final Submission Deadline .....                        | 5        |

|                                                    |    |
|----------------------------------------------------|----|
| 6. AI Usage Rules.....                             | 6  |
| 6.1. AI Integration is Mandatory.....              | 6  |
| 6.2. Permitted and Prohibited Uses.....            | 6  |
| 6.3. The Nexus Log — Mandatory Documentation.....  | 6  |
| 7. Grand Finale.....                               | 7  |
| 7.1. Venue & Attendance.....                       | 7  |
| 7.2. Schedule.....                                 | 7  |
| 7.3. Booth & Demonstration.....                    | 7  |
| 8. Awards & Recognition.....                       | 8  |
| 8.1. Total Prize Pool: RM 10,000.....              | 8  |
| 9. Conduct & Integrity.....                        | 9  |
| 9.1. Original Work.....                            | 9  |
| 9.2. Respectful Conduct.....                       | 9  |
| 9.3. Confidentiality of Judging Criteria.....      | 9  |
| 9.4. Intellectual Property.....                    | 9  |
| 10. Disqualification.....                          | 10 |
| 10.1. Grounds for Team Disqualification.....       | 10 |
| 10.2. Grounds for Individual Disqualification..... | 10 |
| 11. General Provisions.....                        | 11 |
| 11.1. Finality of Decisions.....                   | 11 |
| 11.2. Rule Amendments.....                         | 11 |
| 11.3. Force Majeure.....                           | 11 |
| 12. Contact Information.....                       | 12 |

# 1. Eligibility

## 1.1. Who May Participate

Project Nexus 2026 is open to pre-university, undergraduate, and postgraduate students currently enrolled in Engineering, Artificial Intelligence, Computer Science, or a related technical field at any Malaysian higher education institution.

## 1.2. Team Composition

- Each team must have a minimum of 2 and a maximum of 4 members.
- All team members must be students at a Malaysian higher education institution at the time of registration.
- A student may only register with one team for the duration of the competition.
- Teams may include members from different universities or faculties.

## 1.3. Women in Engineering (WIE) Requirement

### **MANDATORY — WIE Membership**

Every participating team must include at least one (1) female member. Teams that do not satisfy this requirement will not be accepted into the competition regardless of other merits.

Additional WIE incentives:

- Teams with at least 50% female representation or a female-designated Team Lead will receive bonus Diversity Credits applied during Phase 1 scoring.
- Teams meeting the 50% threshold will have exclusive access to mentors who are women leaders in engineering and innovation.
- The female participant who demonstrates the highest overall performance in leadership, technical skill, innovation, and AI collaboration will receive the **“Nexus 2026: Woman Innovator of the Year”** individual award.

## 1.4. Registration Fees

Fees are charged per team and are non-refundable after successful submission:

| Team Category                                                                                                        | Registration Fee (RM) |
|----------------------------------------------------------------------------------------------------------------------|-----------------------|
| Teams with 50% of members holding student membership of an organising society (IEEE MMU, IEM MMU, UTP E3S2, EWB MMU) | RM 20                 |
| Teams consisting entirely of MMU students (non-member)                                                               | RM 30                 |
| All other teams                                                                                                      | RM 40                 |

## 2. Competition Tracks

Each team must select exactly one track during registration. Track changes are not permitted after the Phase 1 submission deadline (26 July 2026).

### 2.1. Track 1 — Healthcare Technology

---

- ❖ **Aim:** To revolutionise smart technology in the healthcare sector, improving the accessibility, efficiency, and personalisation of patient-facing solutions.
- ❖ **Scope:** Solutions may address mental wellness, physical rehabilitation, diagnostic systems, patient monitoring, or emergency response.
- ❖ **Challenge:** Design a solution that monitors, diagnoses, or assists patients in real-life medical situations.

### 2.2. Track 2 — Sustainable Solutions

---

- ❖ **Aim:** To design systems that minimise environmental impact, advance the UN Sustainable Development Goals (SDGs), and promote renewable energy.
- ❖ **Scope:** Smart cities, smart agriculture, waste management, green energy monitoring, and renewable energy generation.
- ❖ **Challenge:** Design a solution that optimises resource consumption and reduces environmental stress caused by human activity.

### 2.3. Track 3 — Global Impact

---

- ❖ **Aim:** To address any critical challenge faced by society through inclusive, scalable, and affordable technology.
- ❖ **Scope:** Open to all sectors including, but not limited to, education, public safety, infrastructure, transportation, and social equity.
- ❖ **Challenge:** Identify a critical problem in daily life experienced by societies across the globe and design a solution to it.

## 3. Phase 1 — Registration & Ideation

### 3.1. Key Dates

| Date         | Milestone                                                                          |
|--------------|------------------------------------------------------------------------------------|
| 15 June 2026 | Official registration opens                                                        |
| 11 July 2026 | Registration closes                                                                |
| 13 July 2026 | Participant briefing — track selection confirmed and competition officially begins |
| 26 July 2026 | Phase 1 Proposal submission deadline                                               |
| 29 July 2026 | Announcement of the top 30 selected teams                                          |

### 3.2. Proposal Submission Requirements

By 26 July 2026, each team must submit both of the following via the official registration portal:

1. A written technical proposal detailing the identified problem, proposed solution, system architecture, and build plan.
2. A 3-minute video presentation summarising the proposal and demonstrating the team's understanding of their concept.

Submissions must be in English. Incomplete submissions (missing either component) will not be evaluated.

### 3.3. Phase 1 Scoring Criteria

All submissions are independently evaluated by the Technical Division against the following weighted criteria:

| Criteria               | Weight | Description                                                                                      |
|------------------------|--------|--------------------------------------------------------------------------------------------------|
| Problem Identification | 25%    | Clarity and significance of the problem identified, and the tangible benefit if solved.          |
| Conceptual Innovation  | 20%    | Originality and efficiency of the proposed solution.                                             |
| Integration Strategy   | 20%    | Clarity of the plan for integrating the solution into the problem context.                       |
| System Design          | 15%    | Quality of visual system diagrams, architecture overviews, and technical drawings.               |
| Improvement Plan       | 10%    | Identification of future growth potential and a clear roadmap for development.                   |
| Team Diversity         | 10%    | Diversity across gender, race, institution, and faculty. WIE Diversity Credits are applied here. |

The top 10 teams per track will be selected to advance to Phase 2. The Organising Committee's selection is final.

## 4. Phase 2 — Selection, Funding & Workshops

### 4.1. Seed Grant

#### Prototype Seed Grant — RM 700 per team

Each of the 30 selected finalist teams will receive a RM 700 prototype seed grant, disbursed on 1 August 2026. Funds must be used exclusively for purchasing hardware components required to build the solution prototype.

#### MANDATORY — Seed Grant Financial Compliance

All receipts for components or services purchased using the RM 700 seed grant must be submitted in a separate PDF file together with the updated proposal. Any leftover funds must be transferred back to the Treasurer via bank details provided at submission. Leftover amounts must align precisely with the receipts submitted. Any purchases not directly related to building the prototype will result in **team disqualification** and **blacklisting** from future IEEE MMU SB competitions.

### 4.2. Workshops & Talks

A series of structured workshops will be held by external speakers from **1–25 August 2026**. Topics will include, but are not limited to:

- Effective use of AI tools in engineering workflows
- Hardware prototyping fundamentals
- Software development for embedded systems
- Presentation and pitching skills

Attendance at workshops is strongly encouraged. Schedules and joining instructions will be distributed to all finalists in advance.

## 5. Phase 3 — Build & Mentorship

### 5.1. Build Period

The primary build phase runs from **26 August to 9 September 2026**. During this period, teams are expected to construct and test their physical prototypes using the seed grant components and AI tools provided.

#### MANDATORY — Physical Prototype Requirement

Every team that advances to the Grand Finale must deliver a physical, functioning hardware prototype. Software-only or purely conceptual submissions will not be accepted at the Grand Finale stage. Prototypes must perform their core function live during the booth demonstration.

### 5.2. Mentorship

Each finalist team is entitled to two reserved mentorship sessions with an industry professional aligned to their track:

- Mentorship Week 1: 26 August – 2 September 2026
- Mentorship Week 2: 2 September – 9 September 2026

All teams receive equivalent mentorship access to ensure fairness. Teams must schedule their slots in advance via the coordinator assigned to their track.

### 5.3. Grand Finale Selection

Following the build phase, each team must submit a final video presentation by **16 September 2026** (see Section 5.4). Based on this submission, the top 5 teams per track will be selected to present at the Grand Finale in September.

### 5.4. Final Submission Deadline — 16 September 2026

#### Deadline: 16 September 2026 (23:59 MYT)

Late submissions will not be evaluated under any circumstances. Teams are strongly advised to submit well before the deadline to avoid technical issues.

By 16 September 2026, each team must push all the following into the GitHub repository assigned to them by the committee:

1. All technical documentation (system design, component list, circuit diagrams where applicable).
2. Source code for any software or firmware components.
3. A final video presentation (maximum 5 minutes) demonstrating the working prototype.
4. The completed Nexus Log (see Section 6.3).

## 6. AI Usage Rules

### 6.1. AI Integration is Mandatory

All finalist teams are required to meaningfully integrate the provided AI tools into their prototype development workflow. The use of AI solely for generating documentation, reports, or presentation slides is **insufficient**. Teams must demonstrate AI integration at the level of:

- Prototype design and system architecture decisions
- Software or firmware algorithm development
- Simulation, testing, or debugging of the prototype

### 6.2. Permitted and Prohibited Uses

| ✓ Permitted                                                  | ✗ Prohibited                                                                                     |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Using AI to assist with code generation and debugging        | Sharing AI access credentials with non-participants                                              |
| Using AI to simulate hardware or system behaviour            | Using AI access for work unrelated to Project Nexus                                              |
| Using AI to assist with schematic or algorithm design        | Attempting to circumvent the sandbox environment                                                 |
| Identifying and correcting AI errors as part of the workflow | Presenting AI-generated output as entirely original work without acknowledgment in the Nexus Log |

### 6.3. The Nexus Log — Mandatory Documentation

Every team must maintain and submit a “**Nexus Log**” as part of their final submission. The Nexus Log is a structured document that demonstrates the team’s active and critical engagement with AI tools. It must include:

1. A detailed breakdown of how AI was used to coordinate project workflow and key design decisions.
2. Specific instances where the team identified AI errors or limitations, and how they manually corrected or overrode the AI output. This demonstrates that the team co-works with AI rather than wholly relying on it.
3. A brief comparative analysis of how the AI-assisted final prototype outperformed or differed from the team’s original concept.

Failure to submit a complete Nexus Log may result in **disqualification** from individual awards and a reduced score in the final judging stage.

## 7. Grand Finale — 26 September 2026

### 7.1. Venue & Attendance

The Grand Finale takes place in person at Multimedia University, Cyberjaya Campus. Attendance in person is mandatory for all members of the 15 finalist teams. Remote participation is not permitted for the Grand Finale under any circumstances.

Teams that fail to appear at the Grand Finale without prior written notice to the General Chair will be disqualified and forfeit all associated prizes.

### 7.2. Schedule

| Time               | Activity                                                                                          |
|--------------------|---------------------------------------------------------------------------------------------------|
| 9:00 AM – 11:00 AM | Public exhibition, career booths, and private interviews with sponsor representatives             |
| 12:00 PM – 1:00 PM | Judging panel walkthrough of each project booth and lunch break                                   |
| 1:00 PM – 3:00 PM  | Final pitches by each team to the judging panel                                                   |
| 3:00 PM – 6:00 PM  | Keynote addresses, announcement of winners, award presentation, closing ceremony, and group photo |

### 7.3. Booth & Demonstration

Each team must set up a designated booth and conduct live demonstrations of their working prototype to both the public (9:00–11:00 AM) and the judging panel (12:00 PM–1:00 PM). All team members must be present and available at the booth throughout the exhibition period.

The prototype demonstrated must be the same prototype submitted in the Phase 3 final submission. Substitutions or significant post-submission modifications are **not permitted** without prior written approval from the Technical Division Chair.

## 8. Awards & Recognition

### 8.1. Total Prize Pool: RM 10,000

#### Team Awards

| Award                                 | Recipient                                       | Prize                               |
|---------------------------------------|-------------------------------------------------|-------------------------------------|
| Grand Champion                        | 1 team selected from across all 3 track winners | RM 1,600 + Champion Trophy + Medals |
| Champion — Healthcare Technology      | 1st placed team in Track 1                      | RM 1,000 + Medals                   |
| 1st Runner-up — Healthcare Technology | 2nd placed team in Track 1                      | RM 750 + Medals                     |
| 2nd Runner-up — Healthcare Technology | 3rd placed team in Track 1                      | RM 550 + Medals                     |
| Champion — Sustainable Solutions      | 1st placed team in Track 2                      | RM 1,000 + Medals                   |
| 1st Runner-up — Sustainable Solutions | 2nd placed team in Track 2                      | RM 750 + Medals                     |
| 2nd Runner-up — Sustainable Solutions | 3rd placed team in Track 2                      | RM 550 + Medals                     |
| Champion — Global Impact              | 1st placed team in Track 3                      | RM 1,000 + Medals                   |
| 1st Runner-up — Global Impact         | 2nd placed team in Track 3                      | RM 750 + Medals                     |
| 2nd Runner-up — Global Impact         | 3rd placed team in Track 3                      | RM 550 + Medals                     |

#### Individual Awards

| Award                                   | Recipient                                                                                                | Prize          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------|----------------|
| Nexus 2026: Woman Innovator of the Year | The female participant demonstrating outstanding leadership, technical skill, innovation, and AI mastery | RM 750 + Medal |
| Nexus 2026: Best Innovator of the Year  | The highest-performing individual participant overall across all tracks                                  | RM 750 + Medal |

## 9. Conduct & Integrity

### 9.1. Original Work

---

All submitted work must be original and created specifically for Project Nexus 2026 during the competition period. The following are strictly **prohibited**:

- Submitting work that was previously developed as part of a coursework, thesis, or other competition.
- Plagiarism of code, schematics, designs, or written content from any source.
- Presenting another team's work as your own.

Violations will result in **immediate disqualification**. Serious cases may be referred to the participant's institution for academic integrity proceedings.

### 9.2. Respectful Conduct

---

All participants, mentors, judges, committee members, and guests are expected to engage respectfully and professionally throughout the competition. Harassment, discrimination, bullying, or unsportsmanlike conduct of any kind will not be tolerated. The Organising Committee reserves the right to **remove any individual** or **team** from the competition for conduct violations.

### 9.3 Confidentiality of Judging Criteria

---

Phase 1 scoring criteria are published in this document (see Section 3.3). Grand Finale judging rubrics are withheld until finalists are briefed prior to the event. This policy ensures fairness across all teams. Participants must not attempt to obtain or share non-public judging information.

### 9.4. Intellectual Property

---

Teams retain full ownership of their ideas, designs, and prototypes. By participating in Project Nexus 2026, each team grants IEEE MMU SB a non-exclusive, royalty-free license to feature project materials (photographs, video demonstrations, project descriptions) in competition-related publicity and reports, with appropriate attribution to the team.

## 10. Disqualification

A team or individual may be **disqualified** from Project Nexus 2026 for any of the following reasons. Disqualification decisions are made by the Organising Committee and are final.

### 10.1. Grounds for Team Disqualification

---

1. Failure to include at least one female member in the team (WIE requirement).
2. Missing the Phase 1 submission deadline (26 July 2026) or the final submission deadline (16 September 2026).
3. Submitting a prototype that does not function as described during the Grand Finale live demonstration.
4. Misuse of the AI sandbox access (e.g., credential sharing, off-project use).
5. Misappropriation of the prototype seed grant funds.
6. Plagiarism, academic dishonesty, or submission of previously developed work.
7. Failure to appear at the Grand Finale without prior written notice to the General Chair.
8. Any conduct deemed unacceptable by the Organising Committee under Section 9.

### 10.2. Grounds for Individual Disqualification

---

1. Harassment or discriminatory conduct directed at any person associated with the competition.
2. Failure to submit the Nexus Log (may result in exclusion from individual awards only).
3. Any attempt to impersonate another participant or committee member.

## 11. General Provisions

### 11.1. Finality of Decisions

---

All decisions by the Organising Committee, Technical Division, and judging panel regarding scores, team selections, track assignments, and awards are final and binding. Written appeals regarding procedural matters (not scoring outcomes) may be submitted to the General Chair within 48 hours of the relevant decision. Appeals regarding subjective judging outcomes will not be entertained.

### 11.2. Rule Amendments

---

IEEE MMU SB reserves the right to amend, clarify, or supplement these rules at any time during the competition period. All registered participants will be notified of material changes via the official communication channels (email and the competition's social media platforms). Continued participation after notification of a rule change constitutes acceptance of the updated rules.

### 11.3. Force Majeure

---

IEEE MMU SB shall not be held liable for any failure to perform or delay in performing obligations under these rules arising from circumstances beyond its reasonable control, including but not limited to natural disasters, public health emergencies, government directives, or venue unavailability. In such events, the committee will communicate revised arrangements to participants as soon as practicable.

## 12. Contact Information

For any enquiries regarding these rules or the competition in general, please contact:

| Role               | Contact Details                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Director           | Izzat Nazarudin<br>MUHAMMAD.IZZAT.MUHAMMAD1@STUDENT.MMU.EDU.MY<br>+60 11-65452811<br><a href="https://www.linkedin.com/in/muhammad-izzat-nazarudin/">https://www.linkedin.com/in/muhammad-izzat-nazarudin/</a>                                |
| Assistant Director | Maanic Hamarays<br>MAANIC.HAMARAYS.MARIMUTHU@student.mmu.edu.my<br>+60 14-337 3112<br><a href="https://www.linkedin.com/in/maanic-hamarays-a-l-marimuthu-131a3831b/">https://www.linkedin.com/in/maanic-hamarays-a-l-marimuthu-131a3831b/</a> |

---

### Project Nexus 2026: AI Innovation Challenge

IEEE Multimedia University Student Branch